TRANSLATIONAL BIOMARKER VALIDATION · OVERSIZED 64×36 IN STRESS TEST · SYNTHETIC DATA

Cell-Free Methylation Fragmentomics for Stage I–II Pancreatic Cancer Triage in High-Risk Cyst Surveillance

Victor K. Sato, PhD[1]; Miriam Alverez, MD[1]

[1]Synthetic Translational Oncology Laboratory (Synthetic Demonstration Study)

KEY EMPIRICAL FINDING

Locked plasma cfDNA fragmentomic score achieved 87.3% sensitivity (96/110) and 92.5% specificity (185/200) in blinded external validation (N = 310).

01 Translational Rationale

CA19-9 lacks sensitivity for early-stage pancreatic ductal adenocarcinoma (PDAC) and Lewis-antigen-negative patients.

- **Assay:** Targeted bisulfite-free methylation + 5'-end motif fragmentomics from 4 mL peripheral plasma.
- **Blinded External Validation:** N = 310 prospective biobank samples (110 Stage I–II PDAC; 200 mucinous/serous cysts & healthy controls).

02 Controls & Limitations

- Pre-analytical hemolysis and acute pancreatitis spikes can elevate background cfDNA yield.
- Prospective asymptomatic screening positive-predictive-value requires longitudinal cohort follow-up.

03 Fragmentomic Risk Score Distribution Across Stages

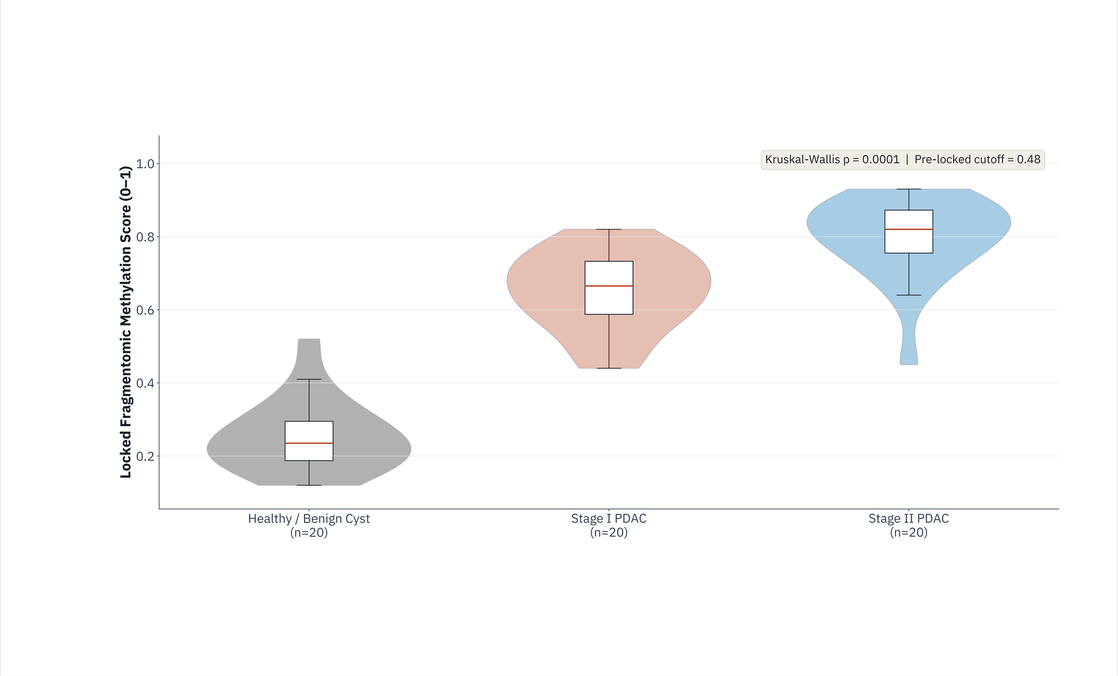


Figure 1. Distribution of locked plasma fragmentomic scores across benign controls and Stage I–II PDAC (synthetic sample distributions).

04 Blinded Validation Contingency Matrix

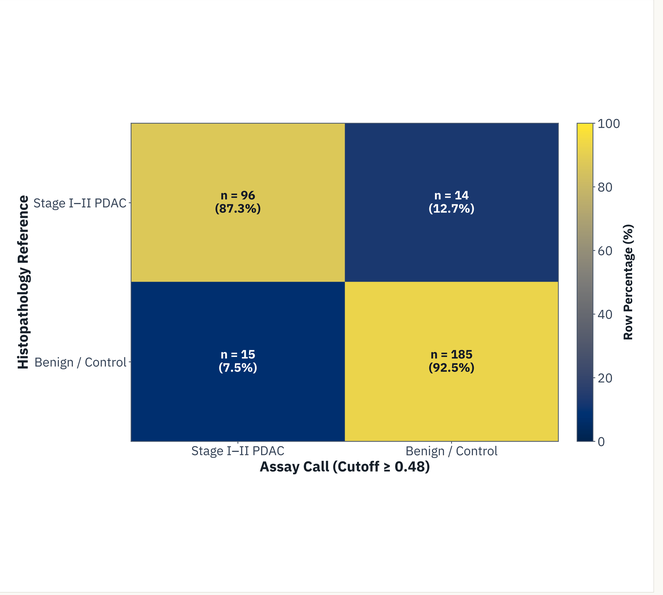


Figure 2. Blinded validation matrix (N = 310): Sensitivity 87.3% (96/110), Specificity 92.5% (185/200).

05 Conclusions

Combined cfDNA methylation and end-motif profiling distinguishes early-stage resectable PDAC from benign pancreatic cysts with >92% specificity.

SYNTHETIC STRESS TEST: Verifies >56-in PowerPoint 50% scale mode + 1:1 full-size 64×36 in PDF + font-fallback audit.